

# TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

## PAPER 2: MATHEMATICAL REASONING

Practice paper · Set 06

**75 minutes**

### INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

**Calculators and dictionaries are NOT permitted.**

**There is no formulae booklet for this test.**

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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**1** Find the value of  $\log_2 40 - \log_2 5 + \log_2 \frac{1}{4}$ .

**A** 1

**B** 2

**C** -1

**D** 5

**E** 3

**F** 0

2 Find the exact value of  $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$ .

**A**  $\frac{\sqrt{3}}{2}$

**B**  $\frac{3}{4}$

**C**  $\frac{\sqrt{3}}{4}$

**D** 1

**E** 0

**F**  $\frac{1}{2}$

- 3** The line  $y = mx + c$  passes through  $(2, 7)$ . If  $m$  is increased by 1 and  $c$  is decreased by 2, the new line passes through  $(2, k)$ . Find  $k$ .

- A** 6
- B** 7
- C** 5
- D** 9
- E** 8
- F** 10

**4** For which of the following values of  $x$  is  $3^x < 2^{x+1}$ ?

**I**  $x = 0$

**II**  $x = 1$

**III**  $x = 2$

**A** none of them

**B** I only

**C** II only

**D** III only

**E** I and II only

**F** I and III only

**G** II and III only

**H** I, II and III

- 5 For a positive integer  $n$ , consider the statement: **\*\*if  $n$  is prime, then  $n$  is odd.\*\*** Which one of the following is true?
- A** The statement is true and its converse is true.
  - B** The statement is true and its converse is false.
  - C** The statement is false and its converse is true.
  - D** The statement is false and its converse is false.

**6** Which of the following geometric series is/are convergent?

**I**  $1 + 0.9 + 0.81 + \dots$

**II**  $3 - 3 + 3 - 3 + \dots$

**III**  $100 + 10 + 1 + 0.1 + \dots$

**A** none of them

**B** I only

**C** II only

**D** III only

**E** I and II only

**F** I and III only

**G** II and III only

**H** I, II and III

7 A student solves  $2\cos^2 x = \cos x$  for  $0 \leq x < 2\pi$ .

**I** Divide both sides by  $\cos x$ :  $2\cos x = 1$ .

**II** So  $\cos x = \frac{1}{2}$ .

**III** So  $x = \frac{\pi}{3}$  or  $x = \frac{5\pi}{3}$ .

How many solutions has the student missed, and where is the first error?

- A** Two solutions missed; the first error is on line I.
- B** No solutions missed; the working is correct.
- C** One solution missed; the first error is on line III.
- D** Two solutions missed; the first error is on line III.
- E** One solution missed; the first error is on line I.

- 8**  $T_n$  denotes the  $n$ th triangular number  $\frac{n(n+1)}{2}$ . A student observes  $T_1 + T_2 = 4$ ,  $T_2 + T_3 = 9$ ,  $T_3 + T_4 = 16$  and conjectures that  $T_n + T_{n+1}$  is always a perfect square. Which of the following proves the conjecture?

**A**  $T_n + T_{n+1} = \frac{n(n+1) + (n+1)(n+2)}{2} = \frac{(n+1)(2n+2)}{2} = (n+1)^2$

**B**  $T_n + T_{n+1} = 2T_n + (n+1)$ , which is a square.

**C** Triangular numbers are always squares.

**D**  $T_4 + T_5 = 25$ , which is a square, so the pattern continues.

**E**  $T_{n+1} - T_n = n+1$ , so the sum is a square.

- 9**  $ABCD$  is a quadrilateral.  $P$ : 'the diagonals of  $ABCD$  bisect each other'.  $Q$ : ' $ABCD$  is a rectangle'. Which one of the following is true?
- A**  $P$  is necessary and sufficient for  $Q$ .
  - B**  $P$  is necessary but not sufficient for  $Q$ .
  - C**  $P$  is sufficient but not necessary for  $Q$ .
  - D**  $P$  is neither necessary nor sufficient for  $Q$ .

**10** Which of the following statements is/are true?

**I**  $\int_{-2}^2 x^3 dx = 0$

**II** The area between  $y = x^3$  and the  $x$ -axis from  $x = -2$  to  $x = 2$  is 0.

**III**  $\int_{-2}^2 x^2 dx = 2 \int_0^2 x^2 dx$

**A** none of them

**B** I only

**C** II only

**D** III only

**E** I and II only

**F** I and III only

**G** II and III only

**H** I, II and III

**11** A student tries to solve  $\sin \theta = \sin 40^\circ$  for  $0^\circ \leq \theta < 360^\circ$ .

**I** Since  $\sin \theta = \sin 40^\circ$ , we have  $\theta = 40^\circ$ .

**II** The sine function has period  $360^\circ$ , so  $\theta = 40^\circ + 360^\circ k$ .

**III** In the given interval the only solution is  $\theta = 40^\circ$ .

Which of the following is correct?

**A** The first error is on line I:  $\sin \theta = \sin \alpha$  does not imply  $\theta = \alpha$ ;  $\theta = 140^\circ$  is also a solution.

**B** The solution is completely correct.

**C** The first error is on line II: the period of sine is  $180^\circ$ .

**D** The first error is on line III:  $\theta = 400^\circ$  should be included.

**E** The first error is on line II: it should be  $\theta = 40^\circ + 180^\circ k$ .

- 12 For real numbers  $a$  and  $b$ , let  $P: 'a = b'$  and  $Q: 'a^2 = b^2'$ . Which one of the following is true?
- A P is necessary and sufficient for Q.
  - B P is necessary but not sufficient for Q.
  - C P is sufficient but not necessary for Q.
  - D P is neither necessary nor sufficient for Q.

**13** Which of the following statements is/are true?

**I** Increasing a quantity by 50% and then decreasing the result by 50% returns it to its original value.

**II** A 10% increase followed by a 10% increase is a 21% increase overall.

**III** If  $A$  is 25% more than  $B$ , then  $B$  is 20% less than  $A$ .

**A** none of them

**B** I only

**C** II only

**D** III only

**E** I and II only

**F** I and III only

**G** II and III only

**H** I, II and III

- 14 Which of the following is a counterexample to the claim **\*\*if  $n$  is a positive integer, then  $2^n + 1$  is prime\*\***?
- A  $n = 8$
  - B  $n = 1$
  - C  $n = 4$
  - D  $n = 3$
  - E  $n = 2$

**15** Three statements are made about a positive integer  $N$ :

- $N$  is a perfect square.
- $N$  is a multiple of 6.
- $N$  is less than 50.

Exactly two of the statements are true. Which of the following could NOT be  $N$ ?

**A** 144

**B** 12

**C** 36

**D** 48

**E** 16

**F** 25

**16** How many positive integers less than 1000 are divisible by neither 2 nor 5?

**A** 399

**B** 600

**C** 500

**D** 400

**E** 450

**F** 300

**17** The function  $f(x) = x^2 - 6x + 13$  is defined for all real  $x$ . Which of the following statements is/are true?

**I**  $f(x) > 0$  for all real  $x$

**II** The graph of  $y = f(x)$  is symmetric about the line  $x = 3$

**III** The equation  $f(x) = 4$  has exactly one real solution

**A** none of them

**B** I only

**C** II only

**D** III only

**E** I and II only

**F** I and III only

**G** II and III only

**H** I, II and III

- 18** For how many integer values of  $n$  with  $1 \leq n \leq 100$  is  $\sqrt{n} + \sqrt{n+1}$  rational?
- A** 9
- B** 1
- C** 10
- D** 0
- E** 2
- F** 100

**19** Claim: the function  $f(x) = x^3 - 3x^2 + 3x$  is increasing for all real  $x$ .

**I**  $f'(x) = 3x^2 - 6x + 3 = 3(x - 1)^2$ .

**II** So  $f'(x) \geq 0$  for all  $x$ , with equality only at  $x = 1$ .

**III** Since  $f'(x) > 0$  for all  $x$ ,  $f$  is increasing.

Which of the following is true?

- A** Line III overstates what was shown:  $f'(1) = 0$ ; but  $f$  is still increasing for all  $x$ , so the claim is true and the proof needs a better justification.
- B** The proof is completely correct.
- C** The first error is on line I.
- D** The first error is on line II:  $f'(x)$  can be negative.
- E** The claim is false:  $f$  has a turning point at  $x = 1$ .

- 20** Let  $P$ : 'the integer  $n$  is divisible by 4' and  $Q$ : 'the integer  $n^2$  is divisible by 8'. Which one of the following is true?
- A**  $P$  is necessary and sufficient for  $Q$ .
  - B**  $P$  is necessary but not sufficient for  $Q$ .
  - C**  $P$  is sufficient but not necessary for  $Q$ .
  - D**  $P$  is neither necessary nor sufficient for  $Q$ .

**END OF TEST**